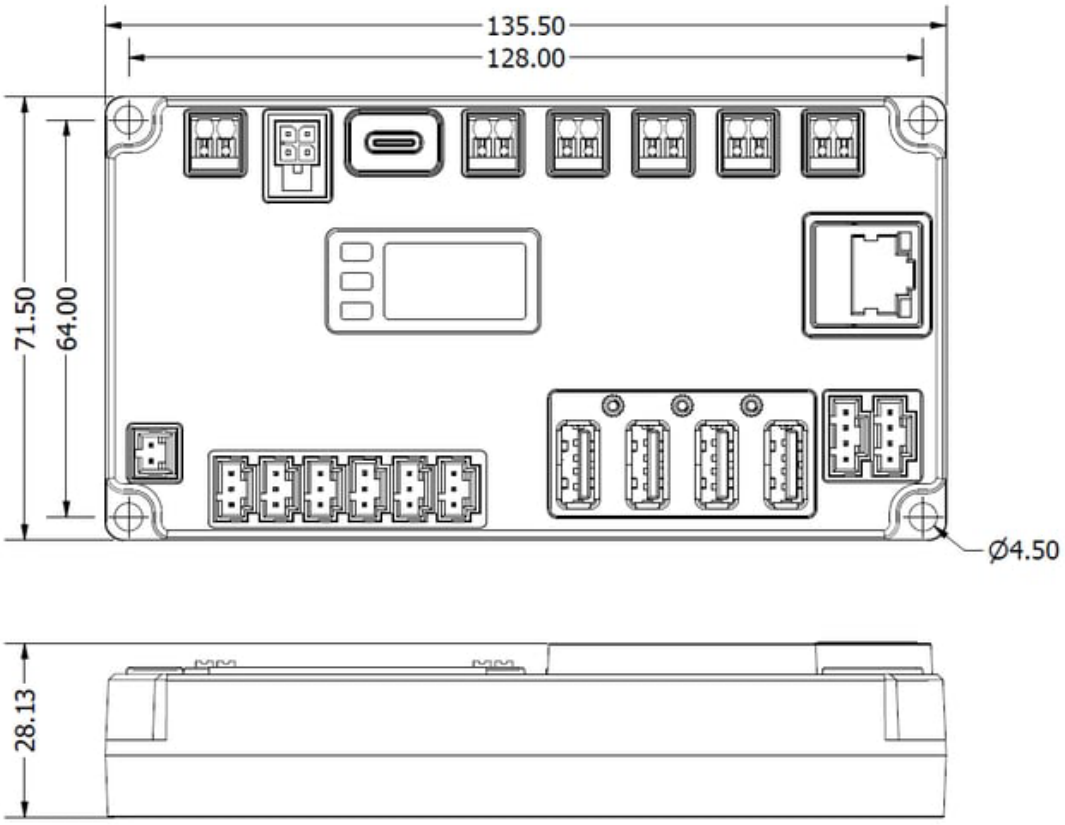
LIMELIGHT SYSTEMCORE - SPECIFICATIONS

This product is currently undergoing alpha testing and is subject to change. All features and metrics marked with asterisks (*) are in-progress or subject to change. Some ratings, such as shock and vibration ratings, are unlisted pending final testing.

6/24/25 – Adjust SIGNAL High Voltage, add mating connector table, add screen fault information

10/1/25 – Fix mating connector part numbers





Physical Characteristics

Weight	215 g (.475 lb)
Dimensions	135.5mm x 71.5mm x 28.13mm (5.3” x 2.8” x 1.1”)
Enclosure	Aluminum Bottom Housing, ABS/PC Top Housing
Conformal Coating	Applied to primary PCB
Operating Temperature Range	(Pending Certification)
Storage Temperature Range	0°C to 70°C*
Operating Humidity	10% RH to 90% RH (non-condensing)*

Powered by Raspberry Pi CM5

The CM5 serves as Systemcore's host processor. It directly handles higher-level interfaces such as USB, Ethernet, I2C, PCIe, and CAN Bus while delegating I/O, display, IMU, and fault monitoring to an RP2350 realtime subsystem.

CPU	Quad-Core ARM Cortex-A76 @ 2.4Ghz (BCM2712)
Cache	512KB L2 Per Core, 2MB L3 Shared
RAM	4GB LPDDR4X-4267
Storage	16GB EMMC
GPU	VideoCore VII @ 800Mhz; GLES & Vulkan Support
PCIe	1x Lane, PCIe 2.0
Wi-Fi / BT	Dual-band 802.11ac (2.4GHz & 5GHz); BT 5.0 / BLE
OS	LIMELIGHT OS; Linux 6.6 with PREEMPT_RT patches

Raspberry PI RP2350 Realtime I/O Subsystem

The host processor loads firmware directly into the I/O subsystem's SRAM on every boot, guaranteeing compatibility between the host and I/O subsystem.

CPU	Dual-Core ARM Cortex-M33 @ 150Mhz
RAM	520KB SRAM
Storage	None

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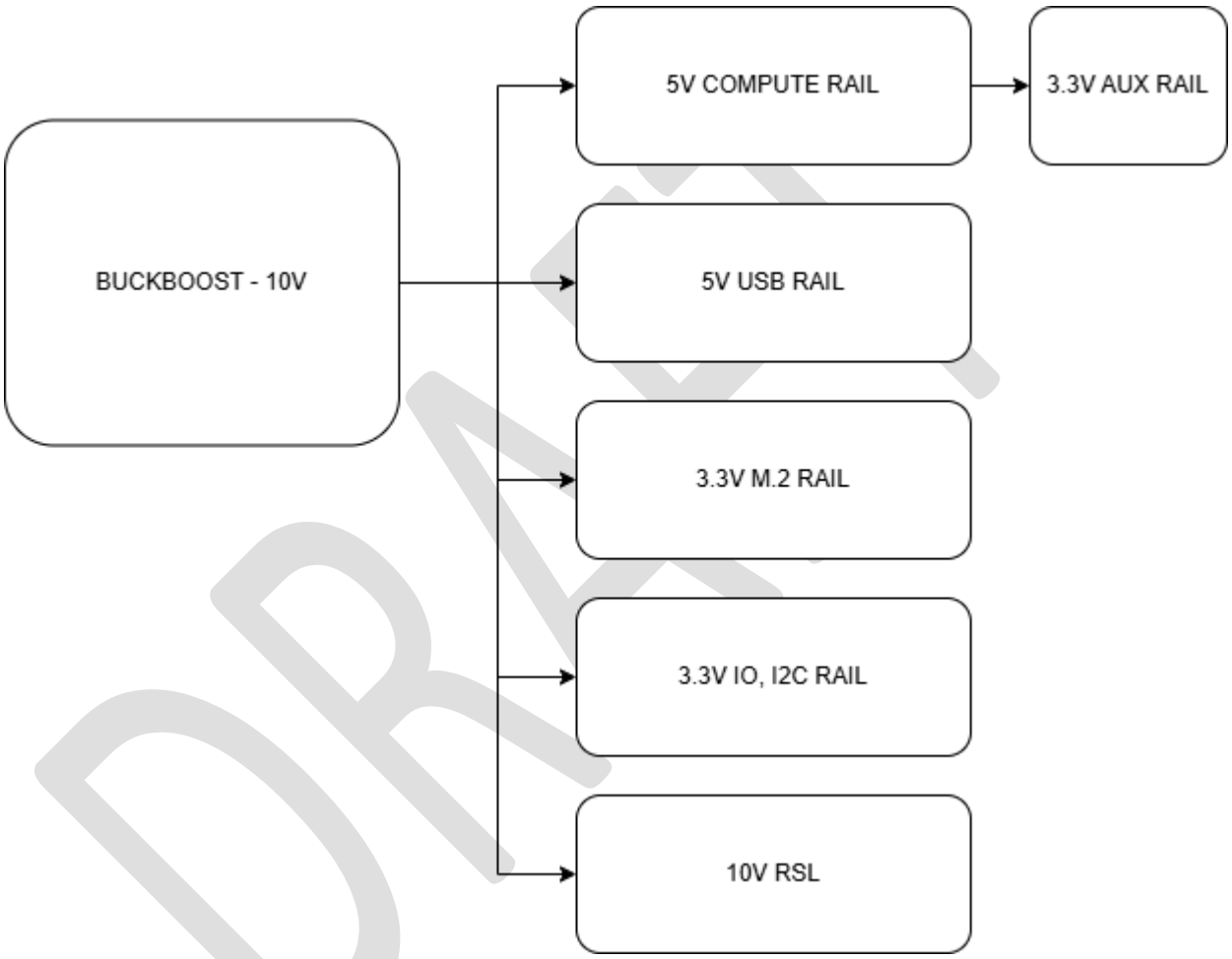
Power Input

Architecture	Buck-Boost
Voltage Range	5V – 26V / 6S Nominal (4.5V - 35V Survivability)
Power Consumption	6W Idle; 40W Max;
Undervoltage Lockout	3.69V*
Brownout Protection Threshold	Configurable. 6V Nominal*
Reverse Polarity Protection	Yes
ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2
Connector	Weidmuller 1825640000
Wire Requirement	18AWG, Ferruled

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Power Rails

All power rails feature overcurrent protection. This is combined with subsystem-level overcurrent protection. Systemcore detects faults and reports them via robot telemetry, the onboard screen, and the STATUS light.



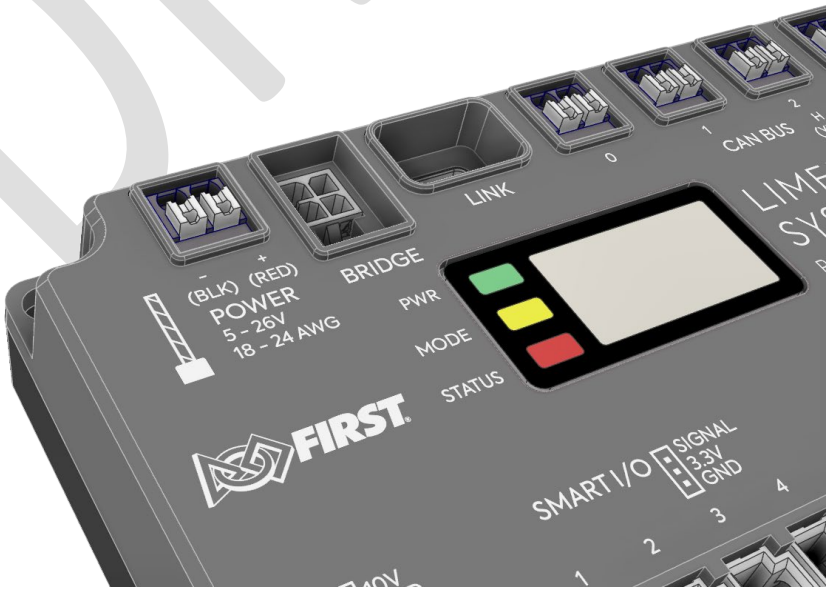
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CAN Bus

Number of Unique Interfaces	5
FD-Capable	Yes
Rate	Up to 8Mbps
Built-in Termination Resistance	120Ω
Overvoltage Protection	+/- 24V Differential; +/- 58V Common Mode
ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2
Connector	Weidmuller 1825640000
Wire Requirement	22AWG, Ferruled

BRIDGE Port

The BRIDGE port (Molex Microfit+ 206832-0401) allows for power input and CAN communications via one cable. It is designed to be used **only** with Motioncore



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I/O Ports

I/O Ports are powered by the PIO subsystem of the onboard RP2350B. I/O Functionality is reconfigurable at runtime.

Function	Digital In/Out, PWM In/Out, Analog In, Addressable LED*, Quadrature Decoding*
Number of Ports	6
ADC Resolution	12bit ADC, 8.75-9bit ENOB
ADC Measurement Range	0.0V - 3.3V
ADC Aggregate Sample Rate	Up to 500 kS/s*
ADC Linearity	99.9%
Pull Configuration	4.7KΩ Pull-Down
VOUT Voltage	3.3V Nominal
VOUT Current Limit	1A Max (Shared between all I/O and I2C Ports)
VOUT ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2
VOUT Overcurrent Protection	I2C and IO VOUT disabled; Fault report; Fault report; Fault cleared on powercycle;
VOUT Overvoltage Protection	24V Max
SIGNAL Input Low Voltage	0.0V Min; .9V Typ
SIGNAL Input High Voltage	2.25V Typ*; 24V Tolerant;
SIGNAL Output Low Voltage @ 7mA	0.05V Nominal

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SIGNAL Output High Voltage @ 7mA	3.27V Nominal
SIGNAL Min Pulse Width	1µs*
SIGNAL Pulse Width Resolution	16 bit Max
SIGNAL Current Limit	10mA Max
SIGNAL ESD Protection	+/- 30kV Air and Contact
SIGNAL Overcurrent Protection	SIGNAL output disabled
SIGNAL Overvoltage Protection	24V tolerant
Fault Reporting	Robot Telemetry*, STATUS LED, OLED Display
Connector	Molex SL 171971-0003

I2C Ports

Speed	100kHz (default), 400kHz (fast mode)
Number of Unique Interfaces	2
VOUT Voltage	3.3V Nominal
VOUT Current Limit	1A (Shared between all I/O and I2C Ports)
VOUT ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2
VOUT Overcurrent Protection	I2C and IO VOUT disabled; Fault report; Fault cleared on powercycle;
VOUT Overvoltage Protection	24V

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SCL/SDA Internal Pullup	Internal Pullups
SCL/SDA Input Low Voltage	0.0V Min; .8V Max
SCL/SDA Input High Voltage	2.0V Min; 24V Max
SCL/SDA Pin Current Limit	14mA
SCL/SDA Pin ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2
SCL/SDA Pin Overvoltage Protection	24V tolerant
Fault Reporting	Robot Telemetry*, STATUS LED, OLED Display
Connector	Molex SL 171971-0004

USB Host Ports (Type A)

USB Host ports accept devices such as USB flash drives, cameras, and expansion devices.

USB Standard	USB 3.0 SuperSpeed
Bandwidth	5 Gbps total upstream (Shared across ports)
Number of Ports	4
VOUT Current Limit	2A (Shared across ports)
Overcurrent Response	USB Subsystem VOUT disabled; Fault report; Fault cleared on powercycle;
ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2

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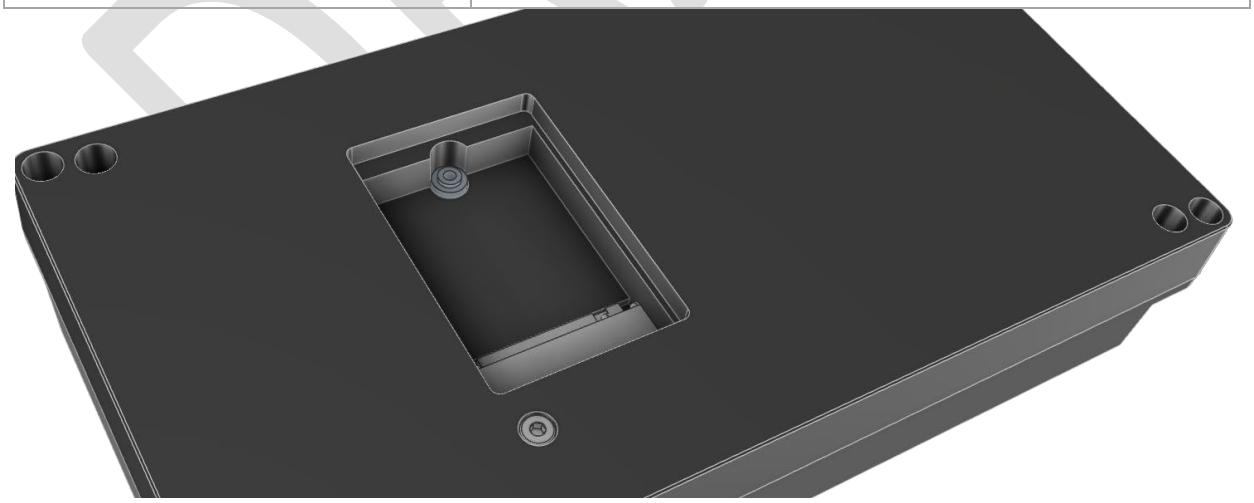
Ethernet Port

Bandwidth	1 Gbps Max
ESD Protection	+/- 30kV Air and Contact
DC Isolation	> 500V*
Autonegotiation Standards	IEEE 802.3 (10BASE-T); IEEE 802.3u (100BASE-TX); IEEE 802.3ab (1000BASE-T)

PCIe M.2 Port

Systemcore must be operated with the included M.2 cover installed.

Connector	A+E Key; Support 30mm length
Rate	PCIe 2.0 (5.0GT/s)
VOUT Current Limit	3A
SIGNAL ESD Protection	+/- 15kV Air and Contact; IEC 61000-4-2



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RSL (Robot Signal Light) Port

The RSL port switches the output of the primary regulator.

VOUT Voltage	10V Nominal
VOUT Current Limit	100mA Max
VOUT ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2
VOUT Overcurrent Protection	RSL VOUT disabled; Fault report; Fault cleared on powercycle
Connector	Molex SL 171971-0002

USB-C LINK Device Port

The LINK port enables cross-platform connectivity over USB-C. This is an alternative interface to the Ethernet and Wi-Fi interfaces. If the LINK port is attached to a device at boot time, Systemcore enters flash mode for recovery OS updates. The LINK port is data-only.

USB Standard	USB 2.0 High-Speed
Bandwidth	480Mbps
ESD Protection	+/- 30kV Air and Contact; IEC 61000-4-2

IMU

Output	400hz Quaternion, Fused Robot Yaw, 3 axis acceleration, 3 axis gyroscope
Accelerometer Full-Scale Range	+/-4G to +/-32G
Gyroscope Full-Scale Range	+/-125dps to +/-4000dps

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OLED Display

Resolution	128x64
Color	Monochrome White
Size	.96”
Information	IP Addresses, Hardware Faults, Heartbeat indicators*, Team Number, Software Version Numbers*

LED Indicators

Power	Solid Green – Normal Operation Off – No power / Reverse Polarity
Mode	Solid Yellow – Robot On and Disabled Blinking Yellow – Robot On and Enabled Off – No power
Status	Solid Red – Uncleared hardware fault; Brownout Protection Enabled*; Display fault; Blinking Red – Display hardware fault* (Image 158); Off – No faults

Brownout Protection

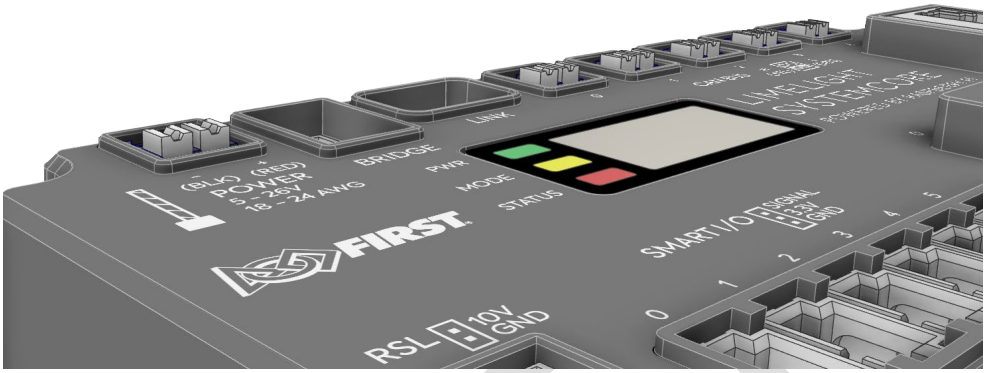
*Brownout protection is disabled in Alpha Image 157

Brownout Voltage	6.3V (Configurable)*
Restoration Voltage	7.5V (Configurable)*
Brownout Protection Behavior	PWM and Digital Outputs Disabled.*

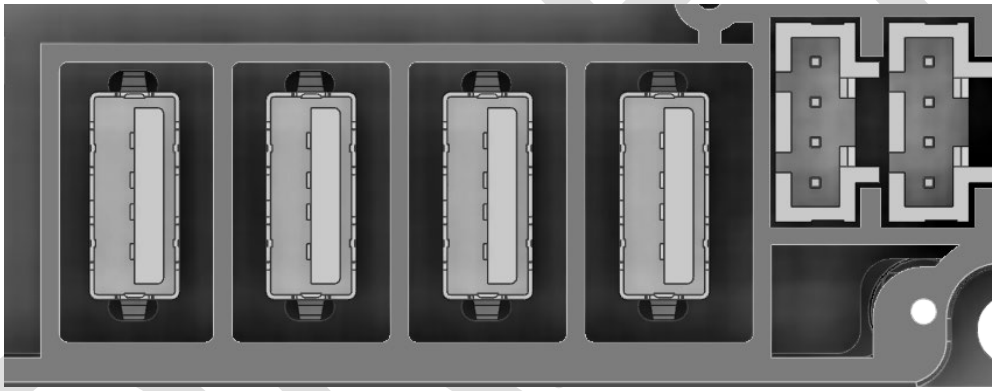
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Debris Mitigation

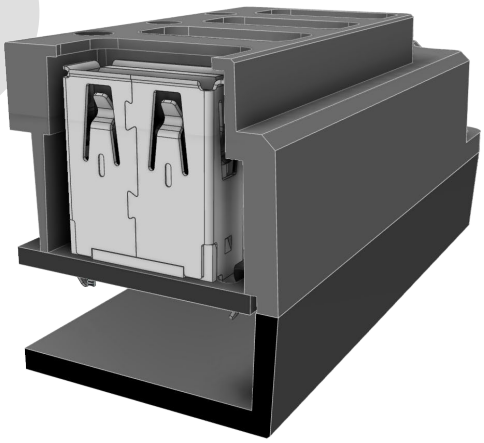
Each connector features raised geometry to prevent debris entry due to vibration.



Each connector or connector group is fully surrounded by the top housing. The top housing makes direct contact with the PCB, as the PCB is installed directly onto the top housing.



The top housing makes contact with the pcb around the edge of the product. This further minimizes the entry of debris into the bottom housing.



Mating Connectors

Port	On-Board Connector	Mating Connector / Terminal	Extension Connector / Terminal
I/O	Molex SL 171971-0003 (GOLD)	50579403 / 16021115(22 AWG GOLD, BAG) 16021111(22 AWG GOLD, REEL)	701070002 / 16020117(22AWG GOLD, BAG) 16020116(22 AWG GOLD, REEL)
I2C	Molex SL 171971-0004 (GOLD)	50579404 / 16021115(22 AWG GOLD, BAG) 16021111(22 AWG GOLD, REEL)	701070003 / 16020117(22AWG GOLD, BAG) 16020116(22 AWG GOLD, REEL)
RSL	Molex SL 171971-0002 (GOLD)	50579402 / 16021115(22 AWG GOLD, BAG) 16021111(22 AWG GOLD, REEL)	701070001 / 16020117(22AWG GOLD, BAG) 16020116(22 AWG GOLD, REEL)
Bridge	Molex Microfit+ 206832-0401 (GOLD)	2064610400 / 206460 SERIES (GOLD ONLY)	2159220401 / 215953 SERIES (GOLD ONLY)

Do not mix gold and tin contacts. Systemcore uses gold contacts for all connectors, so all mating connectors should use gold contacts.

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